

# Pathfinder

From seminar questions  
to research

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**Vincent Danos**

CNRS, ENS

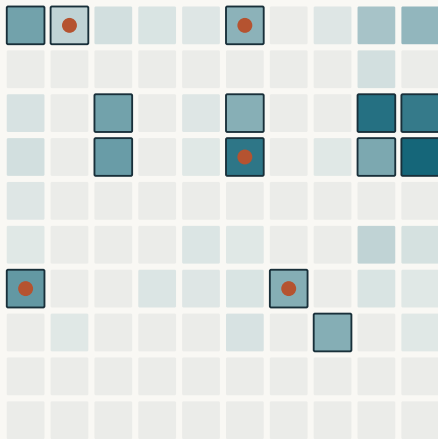
**Elham Kashefi**

CNRS, LIP6  
University of Edinburgh

**William Waites**

University of Edinburgh

[github.com/vd1/pathfinder](https://github.com/vd1/pathfinder)



Ten papers by ten papers.  
Every cell is a possible encounter.

# Convention: mini-abstracts

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## Q mechanism design

### Q4 Incentives to report what one knows

Agents may profit by hiding what they know.  
Rewards can make truthful reporting worthwhile.

### Q7 Sharing resources

Participants compete for limited resources.  
Messages and payments aim to share them well.

### Q3 AI agents in a market

AI agents buy and sell in a simulated market.  
Their performance is compared with humans.

## P systems of cooperating agents

### P6 An outside judge for a team

A team works together; credit is hard to assign.  
An outside AI judges contributions from images.

### P1 Learning about bad outcomes

Learn to act while controlling bad outcomes.  
Estimate losses in the worst fraction of cases.

### P3 A critic for the code reviewer

An AI reviews code; a critic challenges its review.  
The coding agent then revises the program.

# Making connexions ...

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**Bob** “Why use a judge? Could you not ask the agents directly?”

**Alice** “We could ... but wouldn’t they misreport?”

**Bob** “Glad you ask. I have just heard of a mechanism-design paper about that ...”

**Alice** “Interesting!”

## **P6** An outside judge for a team

A team works together; credit is hard to assign.  
An outside AI judges contributions from images.

## **Q4** Incentives to report what one knows

Agents may profit by hiding what they know.  
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# What if we asked across an entire collection?

Mechanism design  $\times$  agent cooperation

|     | P1    | P2   | P3   | P4  | P5  | P6    | P7    | P8   | P9   | P10  |
|-----|-------|------|------|-----|-----|-------|-------|------|------|------|
| Q1  | 1925  | 600* | 300  | 150 | 75  | 1500* | 0     | 75   | 1050 | 1400 |
| Q2  | 0     | 0    | 0    | 0   | 0   | 0     | 0     | 0    | 300  | 0    |
| Q3  | 200   | 0    | 1925 | 0   | 75  | 1575  | 0     | 0    | 3300 | 3000 |
| Q4  | 300   | 0    | 2100 | 0   | 25  | 3150* | 0     | 25   | 1800 | 3575 |
| Q5  | 75    | 0    | 0    | 9   | 0   | 0     | 0     | 0    | 0    | 0    |
| Q6  | 25    | 0    | 0    | 0   | 100 | 50    | 0     | 0    | 600  | 240  |
| Q7  | 2200* | 0    | 0    | 120 | 75  | 150   | 1650* | 0    | 150  | 25   |
| Q8  | 0     | 25   | 0    | 0   | 0   | 200   | 0     | 1650 | 0    | 50   |
| Q9  | 0     | 0    | 0    | 0   | 0   | 0     | 0     | 0    | 0    | 0    |
| Q10 | 0     | 0    | 0    | 0   | 0   | 0     | 0     | 0    | 0    | 0    |



100 possible encounters  
14 investigated pairs  
5 accepted manuscripts

 feasibility  $\times$  gain  
 investigated  
\* internally accepted

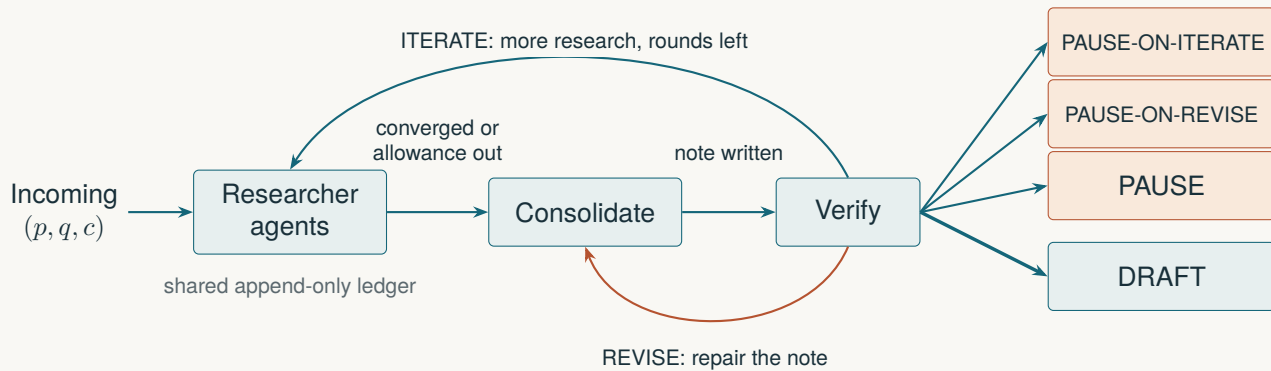
# Scan. Research. Edit.

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The output may be a result, an obstruction, a mistake in either paper, or an account of what is missing to pursue the connexion.

# The research phase



Research question changes after iterations.

Seed  $c$  is a hypothesis to investigate, not a claim the agents must pursue.

Agents are resource-bound.

# Practicalities

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## Recover

Keep artifacts and the ledger.

Resume from the recorded stage.

Retry failed calls within limits.

A transport failure is not a scientific PAUSE.

## Account

Record tokens and elapsed time per call.

Bound calls, rounds and time.

Check budget before admitting work.

An admission guard is not a hard cap on the final bill.

## Reuse

Keep common context at the front.

Append new ledger entries.

Put changing instructions last.

Identical prefixes can be cached across calls.

Prompt context, left to right

|               |                |                             |
|---------------|----------------|-----------------------------|
| Shared papers | Growing ledger | Current call's instructions |
|---------------|----------------|-----------------------------|

Persistent state supports recovery; resource records support control; stable prefixes reduce repeated input cost.

# Q4P6: when can we dispense with the judge?

## Q4 Incentives to report what one knows

Agents may profit by hiding what they know.  
Rewards can make truthful reporting worthwhile.

## P6 An outside judge for a team

A team works together; credit is hard to assign.  
An outside AI judges contributions from images.

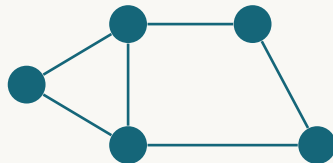
### Seed

Replace the judge with the participants' own reports.

### → Obstacle

Truthful reports can still be insufficient for ranking.

### → Conditional result



overlap  
calibrates  
observers

### Conditions

Informative observations. Structured errors.  
Enough shared comparisons.

**Under these conditions, peer reports identify the model's ranking without an external judge.**

Population-level, conditional application. A model-defined ranking is not automatically causal contribution.



# Q7P1: a tale of two runs

## Q7 Sharing resources

Participants compete for limited resources.  
Messages and payments aim to share them well.

## P1 Learning about bad outcomes

Learn to act while controlling bad outcomes.  
Estimate losses in the worst fraction of cases.

**Shared seed:** adapt resource sharing to participants who care about bad outcomes.

### First run

Risk approximation  
**Error bound**

Mechanism application assumes  
a suitable mechanism exists

### Rerun

Q7's printed payments  
**Gain 9.5; pay 10.5**

Follow-up: repair the payments  
and strengthen the assumptions

The first run bounded the error in estimating risk, but left the resource-sharing rules conditional. The rerun found that Q7 could charge a participant more than they gained, prompting a repair under stronger assumptions.

**The approximation bound survives. Its intended application needed repair.**

# Q3P3: a promising question, an unfinished test

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## Q3 AI agents in a market

AI agents buy and sell in a simulated market.  
Their performance is compared with humans.

## P3 A critic for the code reviewer

An AI reviews code; a critic challenges its review.  
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### Seed: spot a rush to finish

Q3 finds urgency in traders' language just before a trade. P3's reviewers sometimes agree too soon.

**Could Q3's language analysis warn when P3's reviewers are rushing to agree?**

### Shift: test simpler agents

The researchers found that simple random traders could do well under Q3's rules.

This suggested a different question for P3: **does the critic help, or do extra rounds of review explain the gain?**

### Test: compare fairly

Compare a reviewer working alone with a reviewer helped by a critic.

Let both review and repair repeatedly, on the same tasks, with equal computing budgets.

**The required materials were unavailable.**

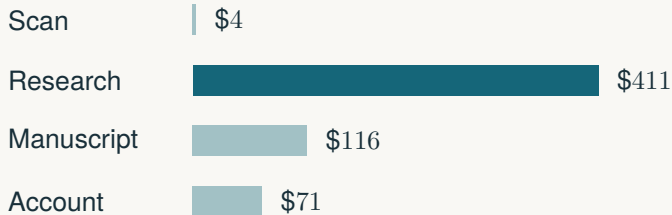
Q3 inspired the test; it did not establish the answer for P3. Neither the seed nor the revised connexion was demonstrated.

Original outcome: PAUSE-ON-ITERATE. Missing evidence, not proof that the critic is ineffective.

# Breadth is cheap. Investigation dominates.

Economics of the 10×10 pilot

**\$601** recorded-token scenario



51.0M recorded input tokens; 1.82M output.  
Flat GPT-6 Standard rates: \$10/M input, \$50/M output.

**43.7 h** campaign span  
Includes pauses and owner continuations.

**8.1 h** summed call time  
Overlapping calls; not wall-clock duration.

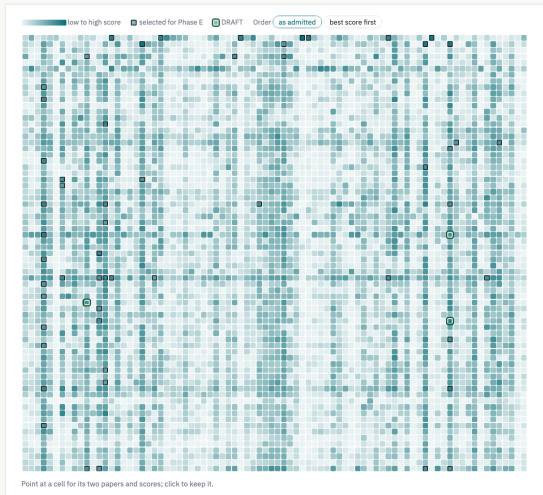
**25 h** illustrative human scan  
Assuming 15 minutes per pair.  
Not a measured human control.

**Scenario, not a bill:** cache usage was incompletely logged; exact repricing is unavailable.

Common tariff, not an observed medium-effort run. Subscription quota is separate. [OpenAI pricing, 16 September 2026.](#)

# Scale-up: the Julien Michel validation experiment

Quantum methods for computational chemistry



$71 \times 82$

5,822 scored pairs

59 pairs selected for research

98 finished threads (51 PAUSE, 44 CONTINUE)

3 DRAFT outcomes

**One pair can suggest several connexions.**

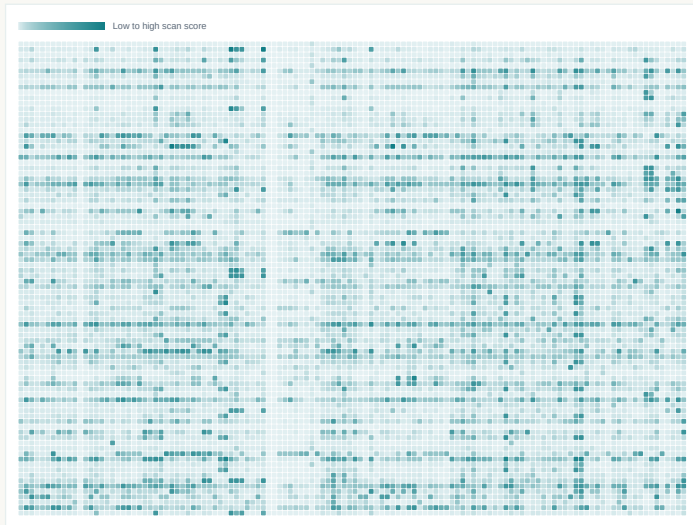
The expert already knows  
one side of the connexion.

Expert assessment becomes a search for directions in their own  
programme.

**Human validation:** do the notes add something new and useful to the expert's research?

# Scale-up in quantum: QSL meets vendor applications

124 vendor-side paper artefacts (columns)



10,912 candidate pairs

100 pairs in the original selection

Fully scored by two judges (Opus 5 and gpt-5.6).

Global rank correlation: **0.740**.



Pathfinder: hypothesis  
generation across the quantum  
software-hardware boundary  
Danos, Kashefi, Waites  
Companion manuscript, 2026

## Later E-prime campaign

147 final notes (144 PAUSE, 3 DRAFT)

Expanded campaign, not just the original 100 pairs.  
Watchboard snapshot: 14 September, 20:42 UTC.

# How much of creativity is connexion?

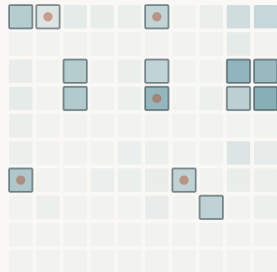
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If agents can make the connexions and also investigate what follows ...

What are the other things humans do in science?

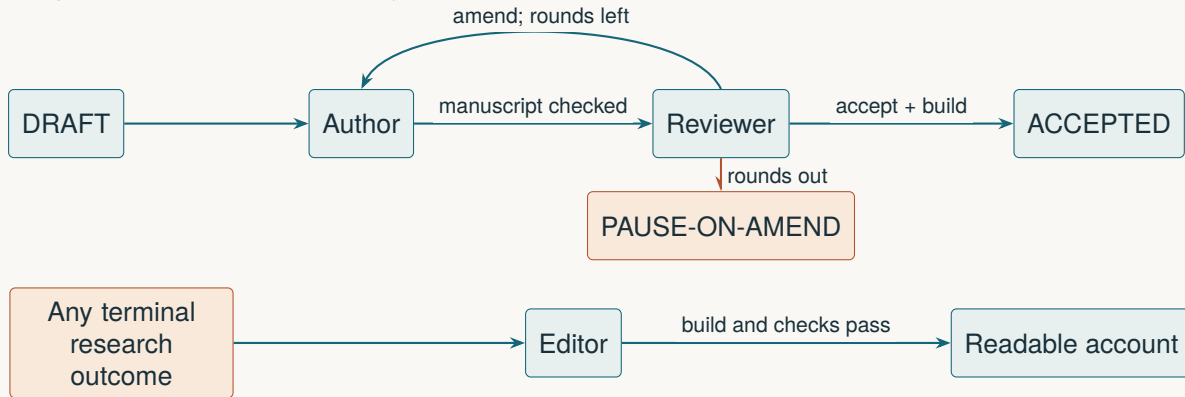
Choosing questions.    Inventing concepts.  
Judging significance.    Applying for funding.

Can agents also help choose which corpora to bring together?



# Backup: editing preserves the research outcome

Two output branches, with different entry conditions



Acceptance refers to the reviewed manuscript version. Neither output branch silently reopens research.

# Backup: what Q4P6 actually assumes

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## **Q4 Incentives to report what one knows**

Agents may profit by hiding what they know.  
Rewards can make truthful reporting worthwhile.

## **P6 An outside judge for a team**

A team works together; credit is hard to assign.  
An outside AI judges contributions from images.

- Truthful reporting under Q4's own incentive and belief assumptions.
- Observers are better than chance, with stable symmetric errors.
- Their errors are conditionally independent on the same latent comparison.
- Overlapping observations contain a calibration triangle and connect onward.
- The agent-comparison graph is connected; the ranking model is specified.

A population-identification result, not a finite-sample guarantee or a theorem that these conditions arise automatically.



# Backup: what the dollar scenario does and does not say

Original 330-call cohort; later runs excluded

| Stage                                   | Calls | Summed call time | Scenario USD |
|-----------------------------------------|-------|------------------|--------------|
| Scan                                    | 106   | 18.3 min         | 3.75         |
| Research + consolidation + verification | 168   | 6.07 h           | 410.56       |
| Manuscript preparation + review         | 38    | 1.23 h           | 116.11       |
| Readable accounts                       | 17    | 28.7 min         | 70.52        |
| Probe                                   | 1     | 4.4 s            | 0.13         |

GPT-6 Standard, per million tokens: input \$10; cache read \$1; cache write \$12.50; output \$50.

Only input and output counters were retained in these receipts. Missing cache and per-request context data prevent exact normalisation.

Scenario applies \$10/M and \$50/M to recorded counters only. Above 272K input tokens, the full request has higher rates. Medium effort is not a multiplier that reconstructs a counterfactual run.

# Backup: why did we pause?

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Proposed review distinction, separate from the stopping mechanism

## PAUSE-PLUS

An actionable continuation exists.

Name the missing material or experiment, the next step, and what its outcomes would establish.

## PAUSE-MINUS

Evidence undermines this direction.

Explain what was ruled out and why, without claiming that every possible connection is exhausted.

An exhausted iteration budget is a stopping reason, not a scientific assessment. Allow “undetermined” too.